

OCR Further Mathematics
Statistics - Chi-squared Tests
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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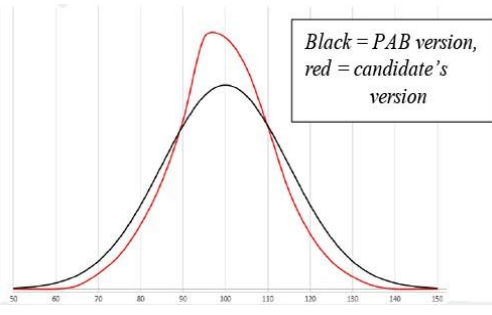
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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.



Question			Answer/Indicative content	Marks	Guidance
1	a		H_0 : Data consistent with $N(100, 15^2)$ H_1 : Data not consistent with $N(100, 15^2)$	B1 (AO1.1) [1]	Allow: "follows $N(100, 15^2)$ " or "can be modelled by". Parameters not needed. No other alternatives seen!
	b		$P(100 \leq X < 110) = 0.2475$ BC Expected frequency = 500×0.2475 $[= 123.754]$ $\frac{(129 - 123.754)^2}{123.754} [= 0.222..., \text{AG}]$	B1 (AO3.4) M1 (AO2.1) A1 (AO2.2a) [3]	Probability needs to be seen Sufficient working to justify AG, needs 123.754 at least
	c		$\Sigma X^2 = 10.5$ $\chi^2(4) = 9.488$ and $10.5 > 9.488$ Reject H_0 . Significant evidence that data is not consistent with $N(100, 15^2)$.	B1 (AO1.1) B1 (AO1.1) M1ft (AO1.1) A1ft (AO2.2b) [4]	Like-with-like comparison needed FT on TS or CV here. Needn't be stated if next line right FT on TS (but not CV) if method correct. Wrong CV, e.g. 5.991: B1B0M1A0. No ft on H_0/H_1
	d	i	E.g. Too few in $X \geq 110$ or in $X \leq 80$, or too many in others, or data truncated, etc	B1 (AO3.5b) [1]	Any relevant point, needn't refer to values of χ^2 "Divide into 5 minute groups": B1. "Data discrete": B0. "The variance" (uncalculated): B0



Question			Answer/Indicative content	Marks	Guidance
		ii		B1 (AO3.3) B1 (AO3.5c) [2]	Deal with aspect identified in (i) Basically correct, areas roughly same <i>Examples:</i> Uses "data discrete" in (i) B0 More below 100, so translate to left B2 More above 110 so translate to right B0 Divide into 5-minute groups B1 Variance changed, areas not equal B0 Data truncated but worse truncation shown
			Total	11	
2	a		$E = 10$ $(19 - 10)^2 / 10 = 8.1$	B1* (AO3.3) B1dep (AO1.1) [2]	Stated or clearly implied Correctly demonstrate 8.1
	b		H_0 : each number equally likely. H_1 : not so $\Sigma X^2 = 23.6$ $\chi^2_{crit} = 21.67$ Reject H_0 . Significant evidence that not all numbers equally likely	B1 (AO2.5) M1 (AO3.4) A1 (AO1.1) M1 (AO1.1) A1 (2.2b) [5]	Both. Oe, e.g. Uniform distribution is good model Add up discrepancies Correct CV Correct first conclusion, ft on their ΣX^2 but not wrong CV Contextualised, not too definite
	c		For 3 and 7: E.g. $p = 0.2$ or 0.195	M1 (AO2.2b) A1 (AO3.5c) [2]	Clearly identify these two numbers Any p with $0.1 < p < 0.5$
			Total	9	



Question			Answer/Indicative content	Marks	Guidance																		
3	a		Expected frequency for Middle/25 to 60 is 4.4 which is < 5 so must combine cells	B1*ft (AO2.4) depB1 (AO3.5b) [2]	Correctly obtain this F_E , ft on addition errors “< 5” explicit and correct deduction																		
	b		<table><tr><td>Early</td><td>Middle</td><td>Late</td></tr><tr><td>29.4</td><td>23.1</td><td>31.5</td></tr><tr><td>26.6</td><td>20.9</td><td>28.5</td></tr></table> <table><tr><td>Early</td><td>Middle</td><td>Late</td></tr><tr><td>0.9918</td><td>0.4160</td><td>2.2937</td></tr><tr><td>1.0962</td><td>0.4598</td><td>2.5351</td></tr></table>	Early	Middle	Late	29.4	23.1	31.5	26.6	20.9	28.5	Early	Middle	Late	0.9918	0.4160	2.2937	1.0962	0.4598	2.5351	B1 (AO1.1) B1 (AO1.1) B1 (AO1.1) [3]	Both, allow 28.4 for 28.5 awrt 2.29, but allow 2.3 In range [2.53, 2.54]
Early	Middle	Late																					
29.4	23.1	31.5																					
26.6	20.9	28.5																					
Early	Middle	Late																					
0.9918	0.4160	2.2937																					
1.0962	0.4598	2.5351																					
	c		H_0 : no association between session and age group. H_1 : some association $\Sigma X^2 = 7.793$ $v = 2, \chi^2(2)_{crit} = 5.991$ Reject H_0 . Significant evidence of association between session attended and age group.	B1 (AO1.1) B1 (AO1.1) B1 (AO1.1) M1ft (AO1.1) A1ft (AO2.2b) [5]	Both. Allow “independent” etc Correct value of X^2 , awrt 7.79 (allow even if wrong in (b)) Correct CV and comparison Correct first conclusion, FT on their TS only Contextualised, not too assertive																		
	d		The two biggest contributions to χ^2 are both for the late session when the proportion of younger people is higher, and of older people is lower, than the null hypothesis would suggest.	M1ft (AO1.1) A1ft (AO2.4) [2]	Refer to biggest contribution(s), FT on their answers to (b), needs “reject H_0 ” Full answer, referring to at least one cell (ignore comments on next highest cells)																		
			Total	12																			



Question		Answer/Indicative content	Marks	Guidance	
4	a	H_0: population frequencies in ratio 9:3:3:1 H_1: population frequencies not in ratio 9:3:3:1	B1(AO1.1) [1]	"Population" can be implied, but <i>not</i> "<u>observed</u> frequencies are in ratio .." <u>Examiner's Comments</u> Many seemed unfamiliar with this type of question and said that the null hypothesis was 'the underlying ratios are <u>not</u> 9:3:3:1'. Some said 'the <u>observed</u> frequencies are in the ratio 9:3:3:1', which is wrong.	Allow "data consistent with model"
	b	Expected frequencies 67.5, 22.5, 22.5, 7.5 $\sum \frac{(O - E)^2}{E} = 4.033 + 4.9 + 2.5 + 0.3 = 11.73$ $11.73 > 11.34$ Reject H_0. There is significant evidence that the results are not consistent with theory	B1(AO3.3) M1(AO3.4) A1(AO1.1) A1(AO1.1) M1ft(AO1.1) A1ft(AO2.2b) [6]	At least 2 correct Correct method for χ^2 awrt 11.7 BC (working not needed) Compare χ^2 value to CV 11.34 ft on TS only. Needs valid TS Conclusion, contextualised, acknowledge uncertainty. <i>Not</i> FT on hypotheses reversed <u>Examiner's Comments</u> By contrast, most could correctly calculate the expected frequencies and the chi-squared statistic, although there were mistakes in selecting the correct critical value. Conclusions were generally well stated.	Allow rounded if seen correct Must see (any) value of χ^2 <i>Not</i> "results <u>are not</u> consistent". FT on "do not reject" needs double negative



Question			Answer/Indicative content	Marks	Guidance	
	c		b as it makes the largest contribution to X^2	B1(AO2.4) [1]	Must refer to test statistic, not “difference”	Or “33” or other clear identification
	d		Increase b (from 3) and <i>either</i> decrease a (from 9) or increase c (from 3), e.g. 7:5:3:1 or 9:5:5:1	B1(AO3.5c) [1]	E.g. 7:4:3:1, 8:4:3:1 or 9:4:4:1 Allow description if b fully correct	Inconsistent to increase c but not
					<u>Examiner's Comments</u> Roughly half the candidates knew that the answer to this question was determined by the biggest contribution to the value of the chi-squared statistic, namely b . Many simply chose a as giving the largest (absolute) difference between observed and expected frequencies.	
					<u>Examiner's Comments</u> This needed to be answered in conjunction with both the values of chi-squared and the raw data. The answer to above part makes it clear that b needs to be changed, and then either a or c . Comparison of observed and expected values shows that the ratio for b is too low, that for a is too high and that for c is too low. So acceptable answers were to increase b and either to decrease a (for instance getting 7:5:3:1) or to increase c as well (for instance getting 9:5:5:1). Again roughly half the candidates gave satisfactory answers.	
			Total	9		

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Question	Answer/Indicative content	Marks	Guidance								
5	<table border="1"> <tr> <td>Expected frequency</td><td>16</td><td>8</td><td>16</td></tr> <tr> <td>$\sum \frac{(O-E)^2}{E}$</td><td>$\frac{4^2}{16}$</td><td>$\frac{(y-8)^2}{8}$</td><td>$\frac{(12-y)^2}{16}$</td></tr> </table> $\chi^2 > 5.991 \Rightarrow$ $1 + \frac{(y-8)^2}{8} + \frac{(12-y)^2}{16} > 5.991$ $\Rightarrow 3y^2 - 56y + 192.144 > 0$ Critical values 4.53 and 14.14 $(0 \leq T) y \leq 4 \text{ or } 15 \leq y \leq 28$	Expected frequency	16	8	16	$\sum \frac{(O-E)^2}{E}$	$\frac{4^2}{16}$	$\frac{(y-8)^2}{8}$	$\frac{(12-y)^2}{16}$	M1(AO3.1a) A1(AO3.3) M1*(AO2.1) depM1(AO3.4) B1(AO1.1b) M1(AO1.1) M1(AO1.1) A2(AO2.2a) (AO3.5b) [9]	Method for expected frequencies Expected frequencies all correct Use $\sum \frac{(O-E)^2}{E}$ $\chi^2 > CV$ Correct CV, ft on cells combined Simplified 3-term quadratic Solve quadratic to obtain CVs www. One number wrong or omitted: A1 <u>Examiner's Comments</u> Many candidates found this question challenging, and in particular many struggled to find the correct expected frequencies. The correct expected frequencies are 16, 8, 16, using the fact that they had to add up to 40. Some thought that the expected frequencies were 8, 4, 8 and hence they combined cells and used the critical value 3.841. Some candidates using one degree of freedom thought that they should use Yates's Correction, but this is not correct; Yates's Correction is not used for goodness-of-fit tests. Other sets of expected frequencies seen quite often were 10.5, 10.5, 10.5 or 12, 6, 12. Many candidates correctly used $\sum \frac{(O-E)^2}{E}$ to obtain a quadratic equation, although some used $\sum \frac{(O-E)^2}{E}$ Candidates often got as Use of cc (Yates?) loses A marks only SC: 8+4+8: M0A0 M1M1B1 M1M1A0A0 Allow = but not $\chi^2 < CV$ 8+4+8: 3.841 (not 5.991) Can be implied Ignore $0 \leq y$
Expected frequency	16	8	16								
$\sum \frac{(O-E)^2}{E}$	$\frac{4^2}{16}$	$\frac{(y-8)^2}{8}$	$\frac{(12-y)^2}{16}$								



Question			Answer/Indicative content	Marks	Guidance
					far as $3y^2 - 56y + 192.144 > 0 \Rightarrow y < 4.53$ and $y > 14.14$ but the values of y have to be integers less than or equal to 28. The final answer $0 \leq y \leq 4$ and $15 \leq y \leq 28$ was rarely seen.
			Total	9	



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Question			Answer/Indicative content	Marks	Guidance											
6		a	Expected frequencies 20, 12, 8; 10, 6, 4 One expected frequency is less than 5	B1*(AO 1.1) depB1 (AO 2.3) [2]												
		b	Mathematics and Physics are the more closely related subjects	B1(AO 2.4) [1]												
		c	H₀: No association between subject and year H₁: there is association between subject & year Combine cells to obtain <table><tr><th colspan="2">Observed</th></tr><tr><td>24</td><td>16</td></tr><tr><td>18</td><td>2</td></tr></table> <table><tr><th colspan="2">Expected</th></tr><tr><td>28</td><td>12</td></tr><tr><td>14</td><td>6</td></tr></table> $\sum \frac{(O - E - 0.5)^2}{E}$ = 0.4735 + 1.0208 + 0.875 + 2.042 = 4.375 > 2.706 Reject H₀. There is significant evidence of association between choice of subject and year of entry	Observed		24	16	18	2	Expected		28	12	14	6	B1(AO 1.1) <
Observed																
24	16															
18	2															
Expected																
28	12															
14	6															



Question			Answer/Indicative content	Marks	Guidance	
		e	Significantly fewer students who enter in Year 12 choose English than would be expected if there were no association.	B1 FT (AO 3.2a) [1]		
			Total	11		
7		a	H_0: "No of pins $\propto$ no of squares" is a good model H_1: "... " not a good model Expected ratios 1 : 8 : 16 : 24 Expected frequencies $\square\square\square\square 2\square\square\square\square$ $16\square\square\square\square\square\square\square\square\square\square 32\square\square\square\square$ 48 Combine cells to give $\square\square\square\square\square\square\square\square\square 27, \square\square\square$ $33, \square\square\square 38; \square\square\square 18, \square\square\square 32, \square\square\square$ 48 $4.5 + 0.03125 + 2.083$ $\chi^2 = 6.61458...$ > 4.605 Reject H_0 Significant evidence that data does not fit model	B1(AO2.5) M1(AO3.3) M1(AO3.1a) M1(AO1.1) A1(AO3.4) M1*(AO1.1) A1(AO1.1b) B1(AO1.1b) M1ft(AO2.2b) A1ft(AO3.5a) [10]	Both hypotheses stated Find ratios Find expected frequencies Combine cells All correct Correct method for calculating χ^2 TS correct, in [6.61, 6.62] Compare with 4.605 Correct conclusion Acknowledge uncertainty	Equivalents acceptable, e.g. in terms of uniform dist FT on wrong TS only
		b	The results for the inner two sectors make the largest contribution to the test statistic and they seem to be aiming at these sectors	depA1ft(AO3.2a) [1]	Contextualised, ft on their calculations, needs M1*	
			Total	11		



Question			Answer/Indicative content	Marks	Guidance
8			$\sum \frac{(O-E)^2}{E} = \frac{1}{N} + \frac{1}{N} + \frac{2^2}{N}$ $\frac{6}{N} = 0.3 \text{ so } N = 20$	M1(AO3.1a) A1(AO1.1) M1(AO1.1) A1(AO1.1) [4]	Correct formula used Correct (O - E) ² values Obtain 20 only
			Total	4	



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Question			Answer/Indicative content	Marks	Guidance										
9			H_0 :gender and car colour not associated H_1 :gender and car colour associated <table border="1"><tr><td>61.6</td><td>78.4</td><td>140</td></tr><tr><td>37.4</td><td>47.6</td><td>85</td></tr><tr><td>99</td><td>126</td><td></td></tr></table> $\chi^2_1 = 6.9^2 \left(\frac{1}{61.6} + \frac{1}{78.4} + \frac{1}{37.4} + \frac{1}{47.6} \right)$ $[= 0.773 + 0.607 + 1.273 + 1.000] = 3.653$ < 3.841 Do not reject H_0 Insufficient evidence of a relationship between gender and car colour	61.6	78.4	140	37.4	47.6	85	99	126		B1(AO2.5) M1(AO1.1) A1(AO1.1) M1(AO3.3) A1(AO3.4) A1FT(AO1.1) M1(AO1.1) A1FT(AO2.2b) [8]	Hypotheses correctly stated (allow “independent”) Correct method for finding E_r All E_r correct Attempt at calculation of χ^2, allow even if no Yates’ Correction Correct value of χ^2, in range [3.65, 3.66] FT Compare their χ^2 explicitly with 3.841 Correct first conclusion Correctly interpreted, need context, acknowledge uncertainty	Yates’ Correction omitted (4.202), reject: can get B1M1A1M1A0A1M1A1 = 7 / 8 FT their 3.653, but not their 3.841
61.6	78.4	140													
37.4	47.6	85													
99	126														
Total				8											